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1 D:\Software\Python\Python311\python.exe E:\PythonProjectsByYear\Year2025\MC-
  PIKAN_Official\experiments\Volterra1D_MCPI.py
2 The network model is cPIKAN
3 Layers: ModuleList(
4   (0): ChebyKANLayer(in_features=1, out_features=10, degree=4)
5   (1): ChebyKANLayer(in_features=10, out_features=10, degree=4)
6   (2): ChebyKANLayer(in_features=10, out_features=1, degree=4)
7 )
8 Total trainable parameters: 600
9 Model on device: cuda:0
10 Start training...
11 Max Epochs: 2000, Learning Rate: 0.001, Epsilon: 1e-20, Optimizer: Adam
12 Training: 100%|██████████| 2000/2000 [00:09<00:00, 222.17epoch/s, Loss=7.
  83e-06, LR=1.0e-03, Best=7.83e-06]
13 Training finished. Best loss: 6.22e-06. Model saved to results/best_model.pth
14 Loss history saved to results/loss_history.npy
15 Training time: 9.82333493232727s
16 Relative loss: 0.0033363972324877977
17 L2 loss: 0.0021229542326182127
18 Parameter containing:
19 tensor([[[ 0.0924,  0.0773,  0.3500,  0.0169, -0.0012],
20          [-0.2785, -0.1536, -0.0391,  0.0074,  0.2384],
21          [ 0.1936,  0.0261, -0.0275,  0.0280, -0.0408],
22          [ 0.3667,  0.1225, -0.1323, -0.3719,  0.0558],
23          [-0.4925,  0.0152,  0.1931,  0.0438, -0.0122],
24          [ 0.2164,  0.2287, -0.1088,  0.2276,  0.3816],
25          [-0.0451,  0.2584,  0.3559,  0.4567,  0.2563],
26          [-0.1058,  0.2184, -0.4464, -0.3320,  0.0082],
27          [ 0.1644,  0.3154,  0.1825, -0.2114,  0.0319],
28          [ 0.3154,  0.2149,  0.2351,  0.1308, -0.0241]]], device='cuda:0',
29       requires_grad=True)
30 Parameter containing:
31 tensor([[[[-1.9263e-02, -5.0413e-02,  1.5464e-03,  4.6620e-02,  7.2333e-03],
32          [-4.4275e-02, -2.4620e-02,  3.7317e-02,  4.4210e-02, -3.3839e-02],
33          [ 2.3087e-02,  5.8483e-02,  2.6143e-02, -1.6236e-02, -1.0754e-02],
34          [ 3.8840e-02,  4.9348e-02, -1.5554e-02, -4.1262e-02, -8.4004e-03],
35          [ 8.3458e-03,  3.2587e-02, -9.5776e-03, -2.1013e-02,  5.7224e-03],
36          [-3.0151e-02,  5.4885e-02, -4.1067e-02, -7.2922e-02,  2.7119e-02],
37          [-1.5405e-02,  1.4974e-01, -3.8078e-03, -1.5438e-01,  1.7738e-02],
38          [-2.3250e-02, -1.2629e-01,  2.8964e-02,  1.5693e-01, -2.2670e-02],
39          [ 3.6866e-02, -2.6630e-02, -3.8004e-02,  4.8816e-03, -1.4643e-03],
40          [-6.5944e-03, -7.3157e-02,  1.0332e-02,  6.9251e-02, -3.0984e-02]],
41          [[ 1.5340e-02,  1.2256e-02,  2.2030e-02, -2.3436e-02, -9.2095e-03],
42          [-4.0060e-02,  1.2972e-02,  1.1274e-02, -1.4671e-02, -6.8172e-02],
43          [-3.9177e-02,  3.7954e-03,  2.3257e-02, -3.3060e-02,  2.2051e-02],
44          [-3.6325e-02, -2.7105e-02, -2.2077e-02, -1.3732e-02, -5.7013e-02],
45          [ 8.6177e-03, -3.0152e-02, -1.7045e-02,  2.9723e-02, -7.5861e-02],
46          [ 1.1505e-02,  6.4619e-03, -3.2708e-02, -1.4930e-02,  5.7370e-02],
47          [-2.0429e-02, -4.7287e-02,  3.3532e-02,  3.0316e-02, -1.7714e-01],
48          [-2.7290e-02, -5.9771e-03,  2.1995e-02,  1.5209e-02,  8.1959e-02],
49          [ 1.8525e-02, -3.5381e-02, -4.9776e-03, -2.7820e-03,  5.2781e-02],
50          [-2.8625e-02,  9.1427e-03, -2.0573e-02,  1.9552e-02, -3.5091e-02]]],
51

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52
53      [[ 2.3994e-03, -2.6917e-02, 1.2360e-03, 3.3606e-02, -2.1420e-02],
54      [-3.0245e-02, -4.1463e-02, 4.0290e-03, 5.8821e-02, -8.0220e-02],
55      [-2.9465e-02, -7.3856e-02, -1.2044e-02, -1.3541e-02, -2.3163e-02],
56      [ 4.5819e-02, 2.9253e-02, -4.9968e-02, 2.6336e-02, -3.1374e-02],
57      [-8.8682e-03, 1.3084e-02, 6.1100e-04, -2.4969e-02, -1.2690e-02],
58      [ 2.2895e-02, 7.4905e-03, -3.5239e-02, -6.7600e-02, 8.8347e-03],
59      [ 1.7826e-02, 6.3107e-03, 2.1217e-02, 1.4805e-02, -2.7962e-02],
60      [-1.0997e-02, -5.0910e-02, 1.8598e-02, 4.4941e-02, -7.5685e-02],
61      [-3.3251e-02, 6.2913e-03, -1.6579e-02, -1.3611e-02, 1.0932e-02],
62      [-6.0025e-03, -5.5490e-03, 2.5971e-03, 3.1344e-02, 9.5560e-03]],
63
64      [[-1.8305e-02, -3.2936e-02, 9.2133e-02, 4.9045e-02, -8.7119e-03],
65      [-8.0795e-02, -1.1221e-02, 1.2035e-01, 4.0380e-02, -1.5349e-02],
66      [-7.2005e-03, 7.2086e-03, -8.1458e-03, 4.7235e-02, 2.8102e-02],
67      [-8.1162e-03, -4.9427e-03, -1.0109e-01, -7.7719e-03, -6.5206e-03],
68      [ 2.5144e-04, 8.5531e-03, 1.9302e-02, -2.3855e-02, -2.9278e-02],
69      [ 1.2614e-02, -2.6295e-03, -9.4282e-02, 1.2713e-03, 5.7455e-03],
70      [-2.6820e-02, 3.2849e-03, -3.9559e-03, -3.6311e-02, 2.0253e-02],
71      [-6.1688e-02, 1.1356e-02, 8.1519e-02, 6.4077e-02, -6.5815e-02],
72      [ 1.1286e-02, 2.4347e-02, -5.0866e-02, -1.8109e-02, -2.3043e-02],
73      [ 1.0874e-03, 2.3020e-02, 7.7651e-02, 1.4628e-02, -6.0222e-02]],
74
75      [[ 3.3246e-03, -4.0112e-03, 3.9238e-02, -2.5614e-02, -1.2099e-02],
76      [-4.5405e-02, 4.0482e-02, 2.9150e-02, -2.8230e-02, -6.3618e-03],
77      [-2.7594e-02, -2.2736e-03, 6.9592e-03, -3.0970e-02, 1.6111e-02],
78      [ 1.5689e-02, 9.4825e-03, -4.3856e-02, 2.0021e-02, 2.6155e-02],
79      [-5.3326e-03, -1.0860e-02, 5.6015e-03, 3.1006e-04, 4.0093e-03],
80      [ 6.2467e-03, -1.9524e-02, 1.4673e-03, -4.0124e-02, -1.6517e-02],
81      [ 4.3616e-02, 2.4521e-02, -4.3461e-02, 2.1094e-02, 1.1389e-02],
82      [-1.6966e-02, 8.2002e-05, 3.1844e-02, -8.3468e-02, -2.5468e-02],
83      [ 1.9012e-02, -6.5755e-03, -3.0975e-02, 2.3479e-02, -2.1993e-02],
84      [ 5.3054e-03, 2.1383e-02, 3.6904e-02, -6.5029e-03, -3.2343e-02]],
85
86      [[ 6.7676e-03, -2.0995e-02, 1.8263e-02, 4.3700e-02, 3.1366e-02],
87      [-3.3604e-02, -1.0821e-01, 1.2783e-02, 8.6192e-02, 4.9363e-02],
88      [-1.9452e-02, -2.1376e-02, 3.1506e-02, 2.2122e-02, 6.7594e-03],
89      [-1.5119e-03, 3.2482e-03, -1.7801e-02, -4.5424e-03, -7.0687e-03],
90      [-5.2846e-02, -3.3227e-02, -6.3284e-02, 3.2174e-02, 4.8556e-02],
91      [ 4.6271e-03, 5.5926e-02, 1.2460e-05, -7.3219e-02, -3.9738e-02],
92      [-7.7897e-03, -1.1400e-01, -2.0344e-02, 8.1172e-02, 3.2231e-02],
93      [-7.6810e-03, -3.1690e-02, 6.5444e-02, 1.3043e-02, 7.8173e-02],
94      [-4.5328e-02, 2.0806e-02, 4.1787e-03, -3.3120e-02, -2.7165e-02],
95      [ 4.9745e-03, -5.5578e-02, 6.0159e-03, 4.5975e-02, 3.7820e-02]],
96
97      [[-2.8265e-02, -8.7830e-03, 8.9194e-03, 1.6973e-02, -9.1044e-02],
98      [-4.2949e-02, -5.7304e-02, 6.0163e-02, -1.1608e-02, -1.5053e-01],
99      [ 2.3356e-02, 5.8580e-03, 2.1285e-02, -1.2443e-02, 1.2961e-03],
100     [ 5.4997e-02, 8.0476e-02, -6.0355e-02, -5.1704e-02, 1.1132e-01],
101     [-2.2769e-02, -3.1688e-02, -3.2648e-03, 6.3487e-02, -5.8443e-02],
102     [ 4.4158e-02, 4.1015e-02, -3.9881e-02, -4.0456e-02, 1.0743e-01],
103     [-3.3581e-02, 3.4233e-02, -1.5365e-02, -4.0324e-02, 5.9781e-02],
104     [-5.0989e-02, -5.3635e-02, 5.7758e-02, 5.8301e-02, -1.0765e-01],

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105      [ 2.0451e-02, 1.2158e-02, 1.5814e-02, 5.4595e-03, 3.9784e-02],
106      [-3.6292e-02, -6.1704e-02, 4.8348e-02, 9.3275e-02, -1.0888e-01]],
107
108      [[-1.0765e-02, 1.5146e-02, 2.9102e-02, -5.5828e-03, -6.1975e-02],
109      [-5.7241e-02, 6.3484e-02, 2.4056e-02, 2.7284e-03, -1.0203e-01],
110      [-2.5460e-02, 4.6544e-03, 4.4194e-02, 1.4504e-02, 1.4274e-02],
111      [-1.6935e-03, -3.9674e-02, -4.6372e-02, -1.4565e-02, 1.0482e-01],
112      [ 2.2880e-02, 3.0995e-02, 4.0453e-03, -4.6886e-02, -8.3208e-03],
113      [ 2.3951e-02, -1.1064e-02, 1.6090e-02, 2.2755e-02, 7.4038e-02],
114      [-2.3226e-02, -3.8688e-02, -6.3645e-02, -3.2412e-03, 1.3538e-02],
115      [-1.8460e-02, 6.2751e-02, 2.7348e-02, -3.3279e-02, -8.2555e-02],
116      [ 4.8899e-03, -1.0063e-02, -3.5529e-02, -1.9165e-03, 4.7181e-02],
117      [-2.2307e-02, 6.0344e-02, 6.3343e-02, -5.0101e-02, -7.9675e-02]],
118
119      [[-3.8360e-02, -2.3153e-03, 1.1612e-02, -1.8222e-03, 6.7026e-03],
120      [-3.2745e-02, -3.3765e-02, 4.8086e-02, 7.8371e-03, -1.0711e-01],
121      [ 1.0005e-03, -1.3534e-03, 1.1391e-02, -2.2419e-02, 9.9079e-04],
122      [ 1.6024e-02, -1.9886e-02, -2.2116e-02, 6.8280e-03, -2.5821e-02],
123      [ 5.2430e-03, 9.9113e-04, 4.0663e-04, -3.9934e-02, -4.3653e-02],
124      [-1.6214e-02, 1.9733e-03, -7.8143e-03, -1.2494e-02, 1.6400e-02],
125      [-2.0945e-02, 3.7637e-02, 4.4548e-02, -5.3210e-02, -8.4523e-02],
126      [-3.4093e-02, -2.2827e-02, 7.9343e-03, -1.9240e-02, -2.1383e-03],
127      [-2.1861e-03, 2.8447e-02, -4.9009e-02, -1.3993e-02, 3.4612e-02],
128      [-8.2947e-03, 3.1822e-02, -1.6288e-02, -5.0255e-02, 8.4803e-04]],
129
130      [[-4.7449e-02, -1.5958e-03, 1.5182e-02, 1.4141e-02, 3.0894e-02],
131      [-1.1883e-02, -5.3381e-02, 4.3885e-02, 5.5280e-02, -2.7950e-02],
132      [-2.3466e-02, -5.6513e-02, 1.6855e-02, 3.3579e-02, -2.1440e-02],
133      [ 5.0450e-03, 8.0367e-02, -3.5789e-02, -5.7204e-02, -3.5642e-02],
134      [ 2.4445e-02, 7.9062e-03, 6.9039e-03, -1.0622e-02, 3.0509e-02],
135      [ 2.7019e-02, 6.0337e-02, -2.4527e-02, -3.5685e-02, -3.5303e-03],
136      [-2.0958e-02, 6.0966e-02, 3.5990e-03, -2.4451e-02, -4.2481e-02],
137      [-1.1149e-02, -5.5874e-02, 1.2744e-02, 8.7003e-02, 6.2459e-02],
138      [ 1.9842e-03, -2.3349e-02, -2.8816e-02, -3.8893e-03, -4.3525e-03],
139      [-3.7526e-02, -5.1567e-02, -5.5895e-03, 5.6752e-02, 4.4982e-02]]],
140      device='cuda:0', requires_grad=True)
141 Parameter containing:
142 tensor([[[[-0.0070, -0.0135, -0.0329, -0.0130,  0.0526]],
143
144          [[ 0.0561,  0.0444, -0.0404,  0.0507, -0.0014]],
145
146          [[ 0.0071, -0.0284, -0.0201,  0.0496,  0.0020]],
147
148          [[ 0.0301, -0.0453, -0.0079,  0.0556,  0.0487]],
149
150          [[ 0.0205,  0.0458,  0.0104, -0.0710, -0.0127]],
151
152          [[ 0.0324, -0.0188,  0.0195,  0.0321,  0.0591]],
153
154          [[ 0.0465, -0.0323, -0.0173,  0.0510,  0.0698]],
155
156          [[ 0.0515,  0.0485, -0.0171,  0.0054,  0.0718]]],
157

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158         [[-0.0312, 0.0094, -0.0374, -0.0116, 0.0221]],
159
160         [[ 0.0314, 0.1153, -0.0229, -0.1198, 0.0851]]], device='cuda:0',
161         requires_grad=True)
162 =====
163
164 =====
165 Layer (type:depth-idx)          Output Shape          Param #
166 =====
167 CPIKANModel                    [1, 1]                --
168 |---ModuleList: 1-1            --                    --
169 |   |---0.weight                |---50
170 |   |---1.weight                |---500
171 |   |---2.weight                |---50
172 |   |---ChebyKANLayer: 2-1      [1, 10]              50
173 |   |   |---weight              |---50
174 |   |---ChebyKANLayer: 2-2      [1, 10]              500
175 |   |   |---weight              |---500
176 |   |---ChebyKANLayer: 2-3      [1, 1]              50
177 |   |   |---weight              |---50
178 =====
179
180 =====
181
182 Total params: 600
183 Trainable params: 600
184 Non-trainable params: 0
185 Total mult-adds (Units.MEGABYTES): 0.00
186 =====
187
188 =====
189
190 Input size (MB): 0.00
191 Forward/backward pass size (MB): 0.00
192 Params size (MB): 0.00
193 Estimated Total Size (MB): 0.00
194 =====
195
196 =====
197
198 进程已结束, 退出代码为 0
199

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